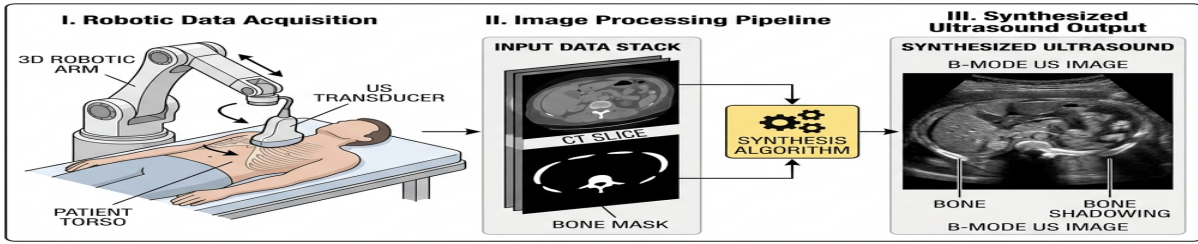


Title: High-Fidelity Real-Time Robotic Ultrasound Simulation via Semantic-Guided Generative Translation

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Robotic ultrasound (US) imaging offers a promising solution for standardized, operator-independent diagnostic scanning and automated medical interventions. However, developing robust control policies requires high-fidelity, computationally efficient simulation environments that can accurately mimic both physical contact dynamics and realistic US image generation. In this work, we present a lightweight, real-time, and motion-adaptive robotic ultrasound scanning simulation framework built in PyBullet, utilizing a clinical curved array transducer probe model mounted on a Franka Panda manipulator. The framework supports both decoupled manual keypress controls and pre-calculated automated trajectory sweeps. To bridge the gap between simulation and real B-mode US scans, we propose a two-channel semantic-guided generative translation pipeline that combines oblique CT slices with bone segmentation masks. The generative translation network is based on a Pix2Pix architecture trained on the UltraBones dataset, utilizing intensity histogram matching to align synthetic slices with the training distribution and resolve domain artifacts. By incorporating highly optimized bilinear interpolation pathways, the framework delivers realistic ultrasound synthesis with sharp acoustic shadowing and specular bone reflections at negligible computational overhead on standard CPU hardware. An active compliance controller is integrated to dynamically modulate probe Z-depth based on estimated penetration distance, enforcing clinical safety limits, preventing probe-torso intersection, and simulating realistic contact dynamics. To ensure smooth interactivity, model inference and simulator visuals are decoupled via background threads, allowing the GUI to run at a smooth 60 FPS. Offline and online evaluations demonstrate that our framework achieves high anatomical consistency and visual fidelity, establishing an accessible, high-performance benchmarking platform for robotic ultrasound control and navigation.